SAMUEL BONNEY

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Research Interests

Nonparametric inference, goodness-of-fit testing, and kernel density estimation, with further interests in time series analysis and biostatistics. My applied work in public health has shaped my interest in how reliable statistical methods are when their assumptions do not hold.

Education

- **Master of Science, Applied Statistics** *Aug 2025 – May 2027 (expected)*
  Illinois State University Normal, IL
  GPA: 3.87/4.00
  Relevant coursework: Applied Probability Models, Applied Stochastic Processes, Statistical Computing, Advanced Linear Algebra, Regression & Time Series Analysis (in progress)

- **Bachelor of Arts, Mathematics and Statistics** *Sep 2019 – Sep 2023*
  University of Ghana Accra, Ghana
  GPA: 3.64/4.00 · First Class Honours
  Relevant coursework: Integration Theory and Measure, Statistical Inference I & II, Multivariate Distributions, Nonparametric Statistics, Regression Analysis, Complex Analysis

Honors & Awards

- **Outstanding Master's Student,** Illinois State University, 2026
- **Kenneth Berk Scholarship,** Illinois State University, 2026
- **JSM Diversity Mentoring Program Award,** American Statistical Association, 2026
- **Fostering Diversity in Biostatistics Workshop Award,** ENAR, 2025 (for the March 2026 workshop)
- **Winner, Statistics and Actuarial Science Quiz Competition,** University of Ghana, 2023
- **Winner, Mathematics Society Quiz Competition,** University of Ghana, 2023

Research Experience

- **Master's Thesis** *Aug 2026 – Present*
  Nonparametric Tilted Density Estimation and Bickel–Rosenblatt Goodness-of-Fit Testing
  Advisor: Prof. Fuxia Cheng, Illinois State University
  - Studying tilted kernel density estimators, where bandwidth and probability weights are chosen by cross-validation, and their use in Bickel–Rosenblatt goodness-of-fit tests.
  - Running simulation studies to compare empirical size and power with tests based on the standard kernel density estimator, and examining the asymptotic behavior of the test statistic.

- **Independent Research Project** *May 2026*
  County-Level Lung Cancer Mortality in the United States
  - Combined CDC Social Vulnerability Index, CDC PLACES, and NCI data for 2,536 counties to study why lung cancer mortality varies across the country.
  - Used spatial regression, LISA cluster analysis, and machine learning (XGBoost with SHAP) to find 232 high-mortality clusters not explained by the measured risk factors; mediation analysis attributed about 50.8% of the smoking–mortality association to COPD.

- **Independent Research Project** *March 2026*
  County-Level Food Insecurity and Chronic Disease Burden in the United States
  - Built a standardized chronic disease burden index for 2,299 counties and modeled its relationship with food insecurity, with sensitivity and spatial autocorrelation analyses.
  - Identified 408 counties with both high food insecurity and high disease burden as candidates for targeted intervention.

- **Research Project** *Jan 2024 – Aug 2024*
  Modeling the Prevalence of Malaria Among Pregnant Women in Ghana Using Explainable Machine Learning
  - Compared machine learning classifiers on Demographic and Health Survey data; k-nearest neighbors performed best (AUC = 0.88), and feature-importance analysis highlighted age and anaemia status as key predictors.

- **Undergraduate Thesis** *Jan 2023 – Sep 2023*

  Machine Learning Models for Identifying Risk Factors of Malaria Among Children Under Five in Ghana
  Advisor: Dr. Gabriel Kallah-Dagadu, University of Ghana
  - Compared five machine learning models, including random forests and neural networks, using the 2019 Ghana Malaria Indicator Survey to identify risk factors for malaria in young children.

Selected Projects

- **BioMed-Scribe: Medical Specialty Classification from Clinical Conversations** *Jan 2026 – Apr 2026*
  - Fine-tuned GPT-2 Medium on 13,492 clinical conversations, training only 7.1% of its parameters, and reached 79.1% test accuracy (macro-F1 0.799) across four specialties; deployed the model as a Gradio web app.
- **Statistical Analysis of Diabetes Risk Factors Among American Adults** *Jan 2026 – May 2026*
  - Analyzed 253,680 CDC BRFSS records in SAS and R using ANOVA, logistic regression, model selection, and sex-stratified models.
- **Gender-Based Analysis of Cardiovascular Disease Risk Factors** *Dec 2025*
  - Fit gender-specific logistic regression models to data on 68,672 adults in Minitab; the combined model reached an AUC of about 0.79.
- **Modeling Life Expectancy in Africa** *Oct 2025*
  - Built a multiple regression model with variable selection and diagnostics; adult mortality and HIV incidence were the strongest predictors.

Teaching & Work Experience

- **Graduate Teaching Assistant**
  Department of Mathematics, Illinois State University Aug 2025 – Present
  - Lead weekly discussion and problem-solving sessions for MAT 113 (Elements of Mathematical Reasoning), MAT 119 (College Algebra), and MAT 120 (Finite Mathematics); grade assignments and exams.
- **Teaching and Research Assistant, Statistical Consulting Unit**
  Department of Statistics and Actuarial Science, University of Ghana Sep 2023 – Aug 2024
  - Led tutorials and review sessions for STAT 223 (Elementary Statistical Methods) and STAT 456 (Actuarial Statistics).
  - Supported faculty research with data cleaning, literature reviews, and exploratory analysis.
- **Enumerator and Data Analyst**
  Ghana Baseline Evaluation Survey Jan 2024 – Feb 2024
  - Collected and analyzed field survey data for a program evaluation, following quality-control protocols.
- **Data Analyst Intern**
  Perception Management International Limited Sep 2022 – Dec 2022
  - Built SQL and Excel pricing models for 50+ due diligence projects and Power BI dashboards that halved weekly reporting time.

Presentations & Conferences

Joint Statistical Meetings (JSM) and Diversity Mentoring Program, American Statistical Association, Boston, MA *Aug 2026*

County-Level Lung Cancer Mortality in the United States: Social Vulnerability, Behavioral Risk Factors, Spatial Clustering, and Unexplained Mortality Hotspots, Research Presentation, Illinois State University *May 2026*

County-Level Food Insecurity and Chronic Disease Burden in the United States: Identifying Priority Counties for Intervention, Research Presentation, Illinois State University *March 2026*

ENAR Fostering Diversity in Biostatistics Workshop, Indianapolis, IN *March 2026*

Application of Machine Learning Models to Identify the Risk Factors of Incidence of Malaria Among Children Under Five Years of Age in Ghana, Departmental Seminar, University of Ghana *Nov 2023*

Skills

- **Statistical Software:** R, SAS, Stata, SPSS, Minitab
- **Programming & Databases:** Python, SQL
- **Data & Visualization:** Power BI, Excel, Geographic Mapping
- **Documentation & Version Control:** LaTeX, R Markdown, Quarto, Git, GitHub

Professional Memberships

Institute of Mathematical Statistics *Sept 2025 – Present*
American Mathematical Society *Oct 2025 – Present*

Leadership & Service

MathEye Impact, Co-Founder / Vice President *2021 – Present*

- Co-founded and help lead a non-profit boarding school for high-achieving students from low-income families across Ghana, mainly the Central Region; the school now serves 275 students.
- Inaugural class achieved the best results in Ghana on the Cambridge International Examinations, with A/A* grades on over 70% of exams taken and 5 students ranked first nationally in their subjects.

Treasurer, Ghanaian Students Association, Illinois State University *2026 – Present*

- Manage the association's budget, financial records, reimbursements, and expenditure tracking, and work with the executive team on event planning.

Solution Space Member, The Mathematics Society, University of Ghana *2020 – 2023*

- Helped fellow students work through challenging mathematics problems and build independent problem-solving strategies.

References

Prof. Fuxia Cheng, Department of Mathematics,
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Dr. Gabriel Kallah-Dagadu,
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